

Knowledge Graph Generation Project - Meeting Specifications and Q&A

Project Overview

Building an efficient and frugal AI pipeline for Knowledge Graph Generation from humanitarian reports, focusing on entity extraction and relationship mapping from ReliefWeb documents.

Questions and Detailed Specifications

Q: Which countries should we focus on for initial development?

A: Focus on two countries with rich data availability:

- **Ukraine** - Due to ongoing war with Russia providing abundant data (last 2 years of reports)
- **South Sudan** - Another data-rich source with ongoing humanitarian situations

Time Frame: Limit to most recent 12 months of documents initially **Minimum Documents:** Target at least 50-100 documents to start (expand timeline if needed)

Q: What document types should we prioritize from ReliefWeb?

A: Focus on 6 specific document types that provide valuable information:

1. **Analysis**
2. **Appeal**
3. **Press Release**
4. **Situation Report (Sitrep)**
5. **UN Documents**

Exclude: Infographics, manuals, guidelines, maps (these won't provide useful extraction data or will bias results)

Implementation: Use ReliefWeb API (not web scraping) and leverage the filter system to get API queries with proper format IDs.

Q: What technical approach should we use for model selection?

A: Multi-model strategy prioritizing efficiency and cost:

For PDF Content Extraction:

- **Option 1:** Classic text extraction (OCR)
- **Option 2:** Vision Language Model (VLM) approach
 - Convert each PDF page to image
 - Use models like Gemini Flash 2.5 for "LLM OCR"
 - Advantage: Can extract context from maps, graphs, visual elements

For Entity Relationship Extraction:

- **Minimum model size:** Llama 70B or equivalent
- **Avoid:** 7B-8B models for complex context understanding (causes hallucinations)
- **Recommended:** GPT-4o Mini minimum, Gemini 2.5 Flash, Llama Maverick

Cost Considerations:

- Gemini 1.5: ~\$1 per 100 documents
- GPT-4: ~\$100 per 100 documents
- Target 100x cost difference analysis

Q: Should we use overlapping window strategy or experiment with multiple approaches?

A: Test multiple strategies on 5 diverse reports:

Strategy 1: Full text extraction (single pass)

- Leverage large context windows (200K tokens $\approx$ 200 pages)
- Advantages: Full context, fewer duplicates
- Disadvantages: May miss details

Strategy 2: Page-by-page extraction

- No overlapping windows
- Better detail capture

Strategy 3: Overlapping window approach

- 50% overlap between windows
- Helps with sections spanning multiple pages
- Cost: 50% more tokens

Validation: Human checking on 5 structurally different reports rather than building evaluation framework initially.

Q: How should we handle duplicate detection?

A: Multi-level deduplication approach:

Entity Level:

- Post-processing clustering for organization names (WFP vs World Food Program)
- Use embedding models for similarity detection

Activity Level:

- Include temporal parameters (dates) to distinguish similar activities at different times
- Example: "WFP delivered food in Gaza (January)" vs "WFP delivered food in Gaza (March)" are separate entities

Relationship Level:

- Remove identical node-relationship combinations

Q: Should we include non-humanitarian activities (military) in extraction?

A: Two approaches:

Approach 1: Extract everything, filter post-processing

- Use binary classification models (humanitarian vs military)
- Leverage tools like GLiNER for single-shot classification
- Allows use of smaller, faster models with cleanup

Approach 2: Large model discrimination

- Use larger models to filter during extraction
- More expensive but cleaner initial output

Recommendation: Start with Approach 1 for scalability.

Q: How important is location mapping accuracy vs processing speed?

A: Prioritize extraction, enhance location post-processing:

Initial Approach:

- Extract location strings as provided by LLM
- Don't expect models to know admin levels initially

Post-processing Enhancement:

- Use geolocation APIs (Google Maps equivalent) for proper admin level mapping
- Target Admin 0 (country), Admin 1 (province), Admin 2 (district) levels
- Secondary priority if time permits

Q: When should we implement validation - during or after extraction?

A: Design evaluation workflows after initial process development:

Human Annotation Approach:

- Take sample reports
- Human annotators create ground truth entities
- Score extracted entities: correct/incorrect/non-existent
- Target metrics: completeness % and accuracy %

LLM Validation Alternative:

- Use state-of-art model (GPT-4o, Sonnet 4) as baseline (99% human equivalent)
- Compare smaller model results against this baseline
- Requires validating the validator (more complex)

Recommendation: Start with human annotation for simplicity.

Q: Should we include map visualization and querying capabilities?

A: Phased approach:

Phase 1 (Core Deliverable):

- Focus on extraction pipeline: PDF → Knowledge Graph
- Output: Neo4j compatible dataset

Phase 2 (If time permits):

- Basic querying examples in Cypher language
- Natural language to Cypher translation tools
- Geographic visualization with lat/long mapping

Minimum Viable Product: Clean ingestion process with structured knowledge graph output.

Key Recommendations and Guidance

Model Selection Strategy

1. **Find the smallest effective model** - Start with reference model (GPT-4o) achieving 99% accuracy
2. **Test progressively smaller models** (Llama 70B → smaller) while maintaining reasonable results
3. **Optimize for smaller models** through better chunking, document parsing, noise reduction

Scalability Considerations

- **Energy Efficiency:** Prefer smaller models for environmental impact
- **Processing Speed:** 70B parameter models significantly faster than trillion-parameter models
- **Privacy:** Smaller models can be self-hosted vs cloud-only large models
- **Cost Analysis:** Include efficiency, speed, environmental impact beyond pure dollar costs

Technical Architecture

Input Requirements:

- Work with PDF files only (no metadata dependency for scalability)
- Extract metadata from PDF content itself
- Consider batch/asynchronous processing for multiple files

Output Requirements:

- Knowledge graph with rich node/relationship attributes
- Source verification: Include page/paragraph references for each extracted relationship
- Temporal data: Include timestamps for activities
- Thematic clustering: Group activities under humanitarian themes (food/nutrition, water, mental health, etc.)

Activity Clustering Strategy

- Use ReliefWeb's thematic areas as clustering categories
- Implement using embeddings or single-label classification
- Post-process extracted activities to assign themes
- Example themes: Food Security & Nutrition, Water Supply, Mental Health

Data Quality Assurance

- **Source Attribution:** Track which page/paragraph generated each relationship
- **Temporal Analysis:** Include date parameters for time-series capabilities
- **Context Injection:** Use report metadata (dates, locations) to enhance extraction accuracy
- **Normalization:** Standardize organization names, location references

Implementation Priorities

1. **Core Pipeline:** PDF ingestion → Entity extraction → Knowledge graph generation
2. **Model Optimization:** Find most efficient model combination
3. **Quality Validation:** Human-annotated evaluation framework
4. **Enhanced Processing:** Geolocation mapping, thematic clustering
5. **User Interface:** Neo4j browser or custom visualization (time permitting)

Success Metrics

- **Completeness:** % of actual entities/relationships captured
- **Accuracy:** % of extracted entities/relationships that are correct
- **Cost Efficiency:** Processing cost per document
- **Processing Speed:** Documents processed per hour
- **Energy Impact:** Model parameter size as proxy for environmental impact